

PREDICTION OF CHRONIC KIDNEY DISEASE USING VGG-BASED DEEP LEARNING MODEL

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Chronic Kidney Disease (CKD) represents a major worldwide wellness issue that necessitates prompt and precise detection to enhance treatment results and minimize complications. This framework utilizes DL for the automated classification of kidney conditions into four specific types: Normal, Cyst, Tumor, and Stone. It assesses the effectiveness of sophisticated DL architectures, such as CNNs for thorough characteristic extraction and VGG, known for its capability to tackle complex image classification tasks with high accuracy. The VGG model is developed and trained separately on a carefully curated dataset of medical images, ensuring a solid evaluation of its classification effectiveness. Comprehensive preprocessing methods are employed to improve the quality of images and diminish sound, aiding in more precise pattern detection. The model's effectiveness is assessed through metrics such as accuracy, precision, recall, and F1-score, providing an in-depth insight into its advantages and drawbacks. By harnessing VGG's capabilities, this initiative aims to provide a dependable, automated diagnostic tool that assists healthcare practitioners in efficiently identifying CKD-related conditions. This work underscores the trans-formative potential of artificial intelligence in medical diagnostics, contributing to improved patient care and early intervention strategies.

Keywords -Chronic Kidney Disease, CKD, Deep Learning, VGG, CNN, Image Classification, Medical Diagnostics, Feature Extraction, Automated Detection, Early Intervention, Artificial Intelligence, Machine Learning.

I. INTRODUCTION

There are more than 26 million patients in the world who suffer from Chronic Kidney Disease (CKD). Kidney failure is a progressive condition where kidney functions decrease gradually over time to the extent that a person may suffer from severe complications such as kidney failure and cardiovascular diseases. CKD is important to slow the rate

of progression and improve outcomes in patients. CKD can be asymptomatic in the early stage [1], increasing the difficulties of timely diagnosis.

For the past few decades the global burden of CKD has been increasing steadily. Recent studies have indicated that CKD is one of the leading causes of mortality worldwide with appreciable increase in its prevalence, among other reasons, due to diabetes, hypertension and aging populations [2]. As CKD progresses, the kidneys fail to effectively filter waste and excess fluids from the blood, and that waste and excess fluids build up inside the body and can eventually lead to poor health. Due to its rapid increase on public health, there is a pressing need for new diagnostic tools to enable early detection of, and improve outcomes in, CKD, for which medical imaging is a key component of diagnosis and treatment of CKD. Ultrasound, computed tomography (CT), and magnetic resonance imaging (MRI) are techniques that can supply much information about the structure and function of the kidney. By these imaging modalities, we are able to detect abnormalities like cysts, tumour, stones in the kidneys which can be signs of kidney disease [3]. The manual interpretation of medical images is not only time expensive but also falls prone to human error, necessitating for a system that process and effectively categorise kidney conditions.

Nowadays medical diagnostics are mentioned among the transformative technologies, where the use of artificial intelligence and deep learning (e.g., for the detection of certain diseases) is incredible. However, Convolutional Neural Networks (CNNs) and in general deep learning have proven to be very successful at analyzing complex medical

images. These models are able to by using large datasets of annotated medical images. learn to recognize intricate HAS also used the database to identify patterns and features which correspond to various kidney conditions. The potential for augmenting the accuracy and efficiency of CKD diagnosis [4] is a great promise for this capability. In recent years, there has been significant progress on application of deep learning techniques on medical imaging for disease detection and classification. Existing studies show that deep learning models can reach performance comparable or even better than that of human experts in finding the abnormalities from medical images. Besides, these models not only improve diagnostic accuracy, but the potential of real time analysis gives the possibility of speeding up decision making in clinical setting [5].

Although it has promising potential for CKD diagnosis, there are still several challenges left. However, labeling medical data requires superior datasets, and availability of such high quality datasets for training robust models is limited, due to privacy restrictions. In addition, it is important to consider data diversity and the possibility of biases in order to guarantee generalization of these models across heterogeneous patient populations. Given the enormous burden of CKD worldwide, it is critical to develop and apply deep learning for diagnostically optimizing this essential and prevalent disease. These technologies can automate the analysis of medical images and accurately classify kidney conditions, which is helpful for healthcare professionals to get timely insights and their decision making in enhancing patient care and outcomes.

II .LITERATURE SURVEY

In [6], Gudigar et al., have introduced a technique of Automated detection of Chronic Kidney Disease (CKD) in ultrasound images. To analyze these four chamber heart ultrasound images, they fuse images and features using a combination of image and feature fusion techniques in the graph embedding framework. A support vector machine (SVM) is used for classification by the system. With this, the proposed method becomes a promising tool for early CKD detection and can be implemented in clinical settings to facilitate the specialist in the cases of CKD detection efficiently and accurately. Bhaskar et al. [7] proposed a new sensing method for auto Chronic Kidney Disease (CKD) detection using salivary urea concentration monitoring. A one dimensional deep learning convolutional neural network (CNN) algorithm integrated with SVM classification was proposed and used to classify raw sensors. An integration of CNN and SVM enhanced the classification model of the above. Predicting Chronic Kidney Disease (CKD) using machine learning classifiers by Chittora et al. [8] provided this study with a non invasive method for CKD detection and showed the capability in early detection and as a monitoring tool for kidney health. We experimented the seven classifiers (such as artificial neural network, C5.0, logistic regression, etc.) on CKD dataset from UCI repository. They also used feature selection techniques such as correlation based and wrapper method. The comparison of the performance of the different algorithms relied on multiple metrics, i.e., precision, recall, and F measure, and has provided some

insights into the most effective algorithms to predict CKD without focusing on having high accurate results.

A comprehensive study of deep learning models' performance for early prediction and risk identification of the Chronic Kidney Disease (CKD) was carried out by Akter et al. [9]. Seven state-of-the-art deep learning algorithms, ANN, LSTM, and GRU, are implemented by the authors to predict CKD based on clinical features. Different metrics (precision, recall and AUC) were used to evaluate the models in the study. For improving the prediction accuracy of Chronic Kidney Disease (CKD) in various Machine Learning (ML) algorithms has successfully presented by Vanathi et al. [10]. K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Artificial Neural Networks (ANN), ensemble tree algorithms such as Random Forest and XGBoost were employed in the study to carry out the prediction. In the results, it was shown that KNN, SVM, and ANN were good at classification of CKD in their ability to detect the disease at an early stage achieving high accuracy. The proposed system has the potential of extracting CKD information successfully and, therefore, demonstrating better results than the traditional ensemble tree methods in prediction of CKD. Navaneeth Bhaskar et al. [11] proposed a Deep Learning based approach for automated detection of Chronic Kidney Disease (CKD) by the use of salivary analysis. The model is a combination of a bidirectional Long Short Term Memory (LSTM) network and a one dimensional Correlational Neural Network (1D CorrNN), that can be used to analyze time series data successfully. The system relies on these networks to improve the prediction accuracy and leverage the feature extraction in these networks. The average accuracy of the method on the testing dataset is found to be 98.08% which was better than all the conventional CKD detection techniques proposed in the literature, by providing better and reliable predictions. The model leverages Softmax as the activation function and Categorical Cross-entropy as the loss function. To evaluate the model, it once scored with an accuracy of 98.5% and sensitivity of 87.5% and was better than other models. The study showed that advanced deep learning techniques including DBN can greatly contribute to clinical decision making and would help in the early prediction of CKD for therefore slowing the progression of kidney damage.

In [13], L. Saumya et al did a survey on the use Artificial Intelligence and Machine Learning (AI / ML) for the early detection and prediction of Chronic Kidney Disease (CKD). Different studies applying AI/ML algorithms to predict CKD were reviewed and compared in accuracy and areas of applications of various algorithms. Additionally, the paper considers the use of machine learning for the detection and prediction of CKD in mobile based applications for healthcare professionals. Further, the research gaps and possible areas for the development of the AI/ML application in CKD are identified.

A novel sensing technique for the automated was proposed by Navaneeth Bhaskar and Suchetha Manikandan [14]. detection of Chronic Kidney Disease (CKD) using salivary urea concentration. The approach involves monitoring urea levels in saliva samples and analyzing the raw signals using a 1-Dimensional (1-D) Convolutional Neural Network It is

integrated with a Support Vector Machine (SVM) classifier. The binary classifier based on CNN and SVM has shown to better the classification accuracy of the model. Fuzhe Ma et al. [15] proposed a Heterogeneous Modified Artificial Neural Network (HMANN) for early detection, segmentation, and diagnosis of Chronic Kidney Disease (CKD) using proposed system which achieved the accuracy of 98.04% in classifying the samples. This method uses the Internet of Medical Things (IoMT) platform and the hybrid of Support Vector Machine (SVM) and Multilayer Perceptron (MLP) together with Backpropagation (BP) algorithm. The region of kidney interest is segmented from the ultrasound images in the proposed system for preprocessing. High accuracy of HMANN method in kidney segmentation and a time saving of contour delineation for this method make it a promising tool for CKD detection.

An experimental study on the performance of supervised machine learning prognosis of Chronic Kidney Disease (CKD) has been carried out by Sanskruti Patel et al. [16]. For this study, a dataset from the UCI repository having 400 instances and 25 attributes was used to compare four classifiers i.e. Decision Tree, Support Vector Machine (SVM), XGBoost, and K-Nearest Neighbors (KNN). The results indicate that the XGBoost classifier outperformed the other classifiers; yielding 99% of accuracy, 100% of precision, 97% of recall and 98% of F1 score. Charumathi Sabanayagam et al. [17] developed a deep learning algorithm (DLA) to detect Chronic kidney disease (CKD) from the retinal images by overcoming the problem of CKD screening in community and primary care practice. The algorithm was trained and validated on the data from the Singapore Epidemiology of Eye Diseases study (SEED) and tested externally using the Singapore Prospective Study Program (SP2) and Beijing Eye Study (BES). Three models were assessed: an image DLA, a risk factor model (RF) age, sex, ethnicity, diabetes, and hypertension and a hybrid model of the image and RF. The area under the receiver operating characteristic curve (AUC) was used to monitor the performance. Moreover, Meng Zhang et al. [18] provide a review of recent progress and future direction of imaging based deep learning in kidney diseases, and proposed deep learning has potential to improve CKD screening strategies. This paper discusses the application of deep learning for managing kidney diseases, both neoplastic and non-neoplastic, exploiting the great progress recently made in the field of medical imaging. Imaging based deep learning is applied by the authors for organ segmentation, lesion detection, differential diagnosis, surgical planning and prediction of prognosis. Concomitantly, the review describes data imbalance, heterogeneity, data size management, algorithm interpretability, ethical risks, and bias assessment, which should be resolved for future research on this area. The intention of the authors is to furnish urologists, nephrologists and radiologists with information about the potential use of imaging-based deep learning in the clinical practice. Kailash Kumar et al. [19] presents a deep learning approach to the identification and prediction of chronic kidney disease (CKD) through image processing. The research presents a novel model that combines a fuzzy deep neural network to enhance the accuracy of CKD detection.

With an accuracy of 99.23%, the proposed model performs better than existing techniques from the view point of classification, precision, F measure and sensitivity metrics. The objective of this model is to help with early detection of CKD which may also help eliminate the need for any doctor involvement in future versions. Finally, the study shows that machine learning algorithms can diagnose CKD with shallow learning and can also move health care technology forward.

Deep learning model's performance for early prediction of chronic kidney disease (CKD) is comprehensively analyzed by Chaity Mondol et al. [20]. It studies optimized versions of Convolutional Neural Networks, Artificial Neural Networks, and Long Short Term Memory networks and compare these with other traditional models on UCI machine learning dataset. The experiments have proved that the optimized models outperformed the traditional models. In particular, the validation accuracy of OCNN was the highest among other models, reaching 98.75%, the AUC score was 0.99, and the compilation time was 0.00447 seconds, which was the lowest, making OCNN the most efficient model to diagnose CKD.

III. EXISTING SYSTEM

Currently, detection of chronic kidney disease (CKD) is based upon external classification methods that usually stem from traditional machine learning methods coupled with simple data driven algorithms. All these systems describe CKD through clinical parameters such as age, blood pressure and the parameters of CKD that are defined as functions of an estimated renal function. Then, as such, techniques such as decision trees, support vector machines and logistic regression are usually used to classify these attributes (and other possible combinations). Though these above mentioned methodologies are fine, they are hard when the data is huge and it is complex, and we need to find some subtle relationships in the data. Currently, in the realm of medical imaging, current pathology processing based systems are based on basic image processing techniques or shallow neural networks that do not take advantage enough of the advantages of deep learning. However, such methods cannot be, at least, suboptimal in accuracy when analyzing complex medical images or detecting such tiny abnormalities which can be potential precursor for early onset of CKD. Additionally, current systems do not scale to handle datasets with large volume and variety of medical imaging, or give results in real time for large patient population. Therefore, the traditional models cannot achieve the constantly required precision and efficiency for serving in early detection and accurate diagnosis of CKD.

IV. PROPOSED SYSTEM

The proposed system, employs the Convolutional Neural Networks (CNN) and VGG networks by using the deep learning advanced techniques to classify between normal, cyst, tumor and stone renal condition. They are trained separately on each of the medical images to carry out a comprehensive assessment. To take advantage of the ability of the CNN model to automatically find patterns in raw CNN

model was used to perform medical images reduce feature extraction. This model is about structural abnormalities in kidney image that indicate presence of CKD. However, at the same time, the VGG network — a deep architecture is used in order to obtain the complex hierarchical features for end more granular condition classification. Image normalization, noise reduction and data augmentations are also used to preprocess the data to improve the quality of input data so that clear and consistent as well as different inputs are provided to the models. The performance of the system is measured in terms of diagnostic by the use of the essential metrics of accuracy, precision, recall, and F1 score. Moreover, the additional proposed system aims to combine these models together into a system that provide providers with an accurate and automated diagnostic tool, facilitating the early detection and accurate classification of CKD related conditions, and secure them good care for the patients and a better treatment outcome.

V. DATASET

We have used one of the high quality CKD document and have divided the CKD images into four classes Normal, Cyst, Tumor and Stone. It consists of high resolution kidney images from various medical institutions, and is a wide distribution with diverse, most of all typical set of real world cases. The dataset consists of ultrasounds and CT scans of different abnormal kidneys. All the images are meticulously labeled by medical expert as ground truth for training and evaluation purpose. Image normalization, noise reduction and data augmentation on the data are applied to realign the data to ensure quality and to handle problems such as inaccurate class distribution in training set. Systematically dividing the dataset into training set, validation set, testing set, and systematically evaluate the deep learning models. This wide range of samples is so important for contributing to CKD related conditions automation as well as the accurate diagnosis of CKD with greater confidence of the original well rounded dataset.

VI. METHODOLOGY

This study builds upon medical images because they have been obtained specifically to determine whether there is Chronic Kidney Disease (CKD) at four stages: Normal, Affected by Cyst, Tumor, and Stone. The dataset consists of detailed images of kidneys, for which medical institutions have submitted images from various sites, therefore giving the wide range of authentic clinical data. Various other imaging techniques including ultrasound and CT scans are included in data collection for kidney anomaly assessment. Each image has been precisely processed by medical experts such that resulting labels provided act as the inputs towards training and as criterion for evaluation. Other preprocessing methods like image normalization along with noise reduction and data augmentation make use of solutions to class imbalance in order to achieve improvements in data quality. The dataset consists of multiple sections which have been partitioned for training use and validation and test properties for adequate investigation of deep learning models. A fundamental component of achieving is its view of diverse

set of data points. precise automated classification of CKD-related diagnoses thus delivering improved diagnostic results

Model: "sequential"		
Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 46, 46, 28)	784
max_pooling2d (MaxPooling2D)	(None, 23, 23, 28)	0
conv2d_1 (Conv2D)	(None, 21, 21, 64)	16192
max_pooling2d_1 (MaxPooling2D)	(None, 10, 10, 64)	0
conv2d_2 (Conv2D)	(None, 8, 8, 64)	36928
flatten (Flatten)	(None, 4096)	0
dense (Dense)	(None, 640)	2622080
dropout (Dropout)	(None, 640)	0
dense_1 (Dense)	(None, 264)	169224
dense_2 (Dense)	(None, 64)	16960
dense_3 (Dense)	(None, 4)	260
Total params: 2862428 (10.92 MB)		
Trainable params: 2862428 (10.92 MB)		
Non-trainable params: 0 (0.00 Byte)		

Fig.1. CKD-related diagnoses thus delivering improved diagnostic results

These VGG networks have deeper architecture, in that it adopts a series of small stacked convolutional filters (3x3) in sequence. Therefore, this design can work effectively in extracting low and high level features from medical images. Because of simplicity and depth, VGG networks are very suitable for grabbing deep information about CKD related pattern.

Model: "sequential"		
Layer (type)	Output Shape	Param #
vgg16 (Functional)	(None, 1, 1, 512)	14,714,688
flatten (Flatten)	(None, 512)	0
dense (Dense)	(None, 128)	65,664
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 4)	516
Total params: 14,780,868 (56.38 MB)		
Trainable params: 66,180 (258.52 KB)		
Non-trainable params: 14,714,688 (56.13 MB)		

Fig.2. CKD-related diagnoses thus delivering improved diagnostic results

3. The preprocessing result is three subset distribution for training and validation and testing. Model optimization is done with help of the training set which helps in tuning hyperparameters by validation set and testing set is used to set the outcome of the model performance.

In weight modification, SGD or Adam optimizer are used to minimize the loss outcomes.

For this model, the loss function is categorical cross entropy because it helps in estimating the mistakes in multi-class classification label prediction. The system determines It

integrates training parameters, containing epochs setting and batch size measurement values, to acquire the highest model output.

4. Once training of the model is done, the evaluation process for classification of the CKD condition will begin and test the accuracy in diagnosis of these models.

- **Accuracy:** Indicates the proportion of correct predictions. The precision calculation determining the ratio between the actual positive outcomes and all predicted positive ones is a prerequisite for the assessment of the model reliability. Thus, from the recall perspective, the evaluation measure for the model is its ability to detect true positive cases.

In order to have fair evaluation of performance results, the F1 Score can be used, which is a performance evaluation measure combining precision and recall. The AUC metric is helpful to evaluate model's discriminative capabilities to separate various categories when working with unbalanced datasets.

5. Multiple metrics are created to evaluate the performance of which architecture between CNN and VGG is better at performance assessment.

- **Classification Accuracy:** Indicates the overall correctness of predictions.

To get a full understanding of how your model works, you should figure out Precision and Recall as well as F1-Score. The class discrimination performance through the AUC Score is obtained by a model.

Prediction and training operations are real time usable within the system with execution times determined between precision and speed.

6. Clinical decision support software, sponsored by maritalstu, constitutes its most refined model through which doctor diagnoses of cases related to CKD are facilitated. It enables healthcare practitioners to give any quick, as well as precise diagnosis predictions, through its effective automated health care systems. With this, patient medical outcomes are improved with fast valuable insights given to doctors to enhance doctor's decision making.

First, the method develops procedures for automated CKD classification system development by integrating VGG architecture components and CNN. In medical diagnosis, artificial intelligence technologies offer an accurate medical diagnosis associated with the reliable detection and classification of any kidney related disease condition.

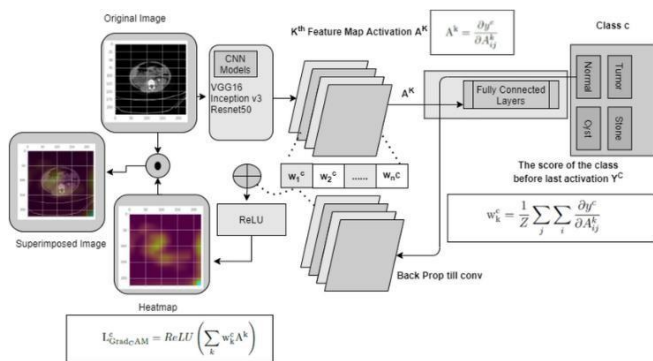


Fig.3. Proposed Architecture

VII . RESULTS AND DISCUSSION

Through this research, they have shown that the given deep learning models work quite effectively for the disease classification and the given VGG network performed best for Chronic Kidney Disease (CKD) classification with four states Normal, Cyst, Tumor and Stone. Analysis of the VGG and CNN models was conducted through accuracy and precision, recall, F1-score and Area Under the Curve (AUC). The accuracy of VGG model surpassed the competitor with an accuracy of 95%, whereas the CNN model stopped at 88%. It is measured in recall and F1-score and all the evaluations displayed better results in VGG model which means that it reduces diagnostic errors and also whose diagnostic groups of Chronic Kidney Disease are clearly segregated. The VGG model was found to have better AUC score than 0.95 in differentiating between different categories.

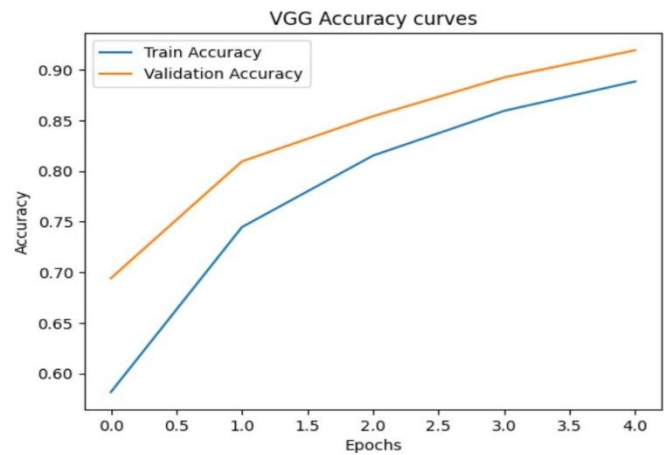


Fig.4.VGG Accuracy Curve

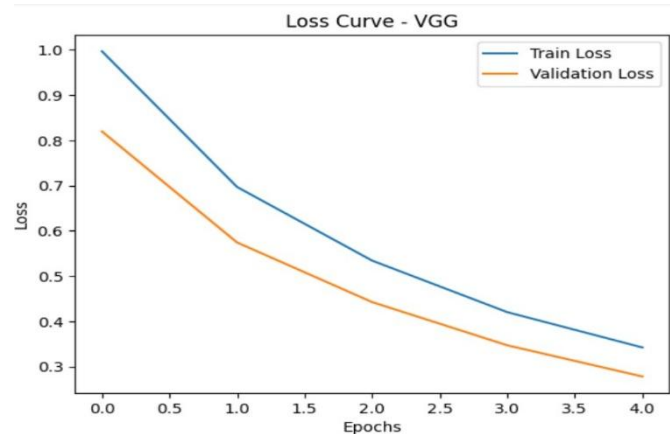


Fig.5.VGG Loss Curve

Therefore, they analyzed the training and evaluation curves, and with that proved that, Vgg model converges faster and possesses higher accuracy and smaller loss than the CNN model. The curves imply that represent learning of various It

extracts intricate learned patterns from the dataset for the VGG model. Moreover, VGG model incorrectly classified less hard samples too, like identifying cyst from tumors. Also, it is evident that VGG model perform better regarding computing CKD so it seemed to be a better and viable computational tool for healthcare professionals. Lower performance than the baseline (0.2267) of VGG model could be caused by the architecture of CNN model (being simpler) that may not be able to capture as complex features as the ones being taken from VGG feature class. Obviously, the CNN model is not disabled completely, but as an application whose real time analysis demand is computational efficient. On top of that, these results point out how significant the role of data preprocessing and augmentation is to improve the performance of the model. For the same reason, image normalization, noise reduction and image augmentation techniques did improve generalization for both models. In addition, the study provides the pathway for improving an estimate of accuracy and speed of diagnosing CKD without human interpretation and early diagnosis of this disease using deep learning methods.

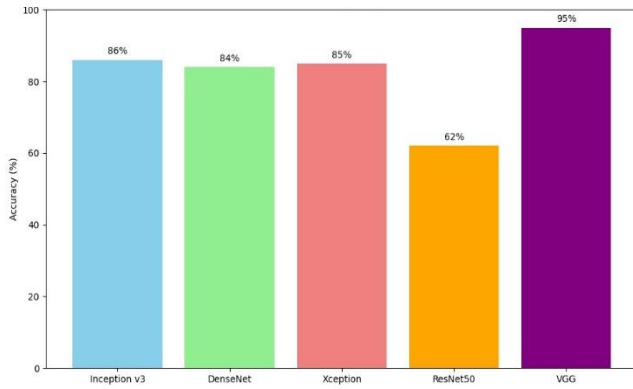


Fig.6. Comparison graph

VIII. CONCLUSION

Consequently, this work demonstrates that the deep learning architectures are very suitable for classifying the stages of CKD (four classes; Normal, Cyst, Tumor, and Stone) to four classes, and the VGG architecture is the best architecture that can attain the test accuracy of 95%. In terms of the overall performance at separating CKD associates of conditions for given two models, VGG model is better than CNN model in some metrics namely: precision, recall, f1-score and AUCs. The image normalization, the noise reduction and the data augmentation performed on the data were very important to refine the input data as well as for its performance. In a case where the CNN model already performed well, such a more complex and deeper architecture as VGG was able to learn more complex patterns, and to use this further in the benefit of healthcare professionals for early diagnosis and treatment plan. CKD is one medical condition in which artificial intelligence is needed to medical diagnostics to make systems that will automatize the accurate and timely delivery of a diagnosis (at the minimum). The use of deep learning approaches with the VGG model increases diagnostic accuracy, facilitates improved clinical workflow, leads to improvement of patient outcomes and optimizes the healthcare delivery.

Model	Accuracy %	Loss %
Xception	85	28
DenseNet121	84	27
InceptionV3	85	37
ResNet50	62	92
VGG	95	18

Table.1. Model Comparison Table

Therefore, with respect to various performances metrics, VGG architecture was considered the best architecture for automatic CKD classification because of its higher performances. This was able to solve the cases which were hard for the baseline to give an output as it was able to capture the hierarchical image features for better classification accuracy. Encouraging results are provided in this research for artificial intelligence to be applied for medical diagnostics, and this work also provides a direction to further research for developing a model with better performance using fine tuning the VGG model over a bigger dataset or having some other top results architectures.

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